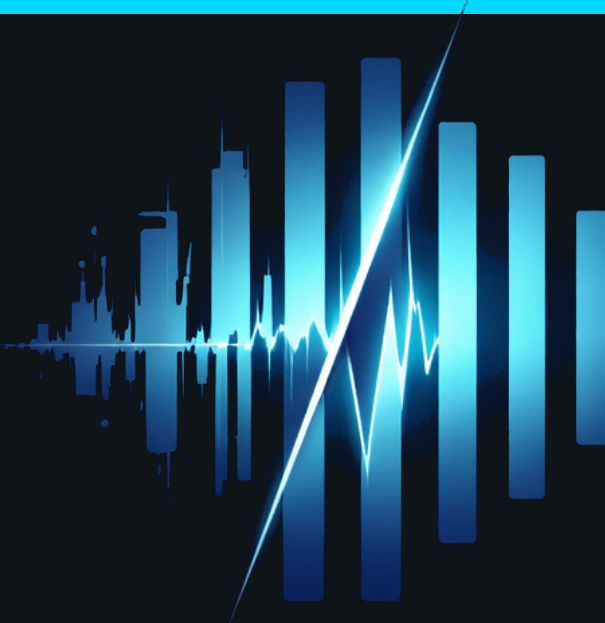




SKIPJACK 2.0

Product-Line Technical Specification

Non Sequitur Publishing

v0.1 - 2026-06-20 - Confidential - prepared for Ologos partners

Architecture in one paragraph

SKIPJACK is one above-Layer-7 custody + freshness **substrate** (the core) with ten product lines riding on it. The platform-OS is uniform - Proxmox + Debian + Docker. Zero-trust, telemetry, and canon are enforced once at the substrate and inherited by every line. There is one build lane (no commercial/federal fork); the distinction is a compliance wrapper applied at sale. Build sequencing runs W0 Foundation, W1 Lead, W2 Integrate, W3 Horizon, PoC-first, over eight quarters. Capabilities are non-exclusive: each has one primary line and may be shared into others, so the shared core fans out across the portfolio.

Each section gives the charter, substrate relation, mapped product, build wave, technical approach, and the capability set (primary + shared).

PL-1 — Custody Bus & Edge-AI Substrate

Substrate: IS the substrate (keel)

Maps to: SJ-01 / FUNDAMENTUM

Wave: W0 Foundation

Technical approach

The keel. An above-Layer-7 mission-state custody layer between edge collection/consumers and transport. It serves the capability locally when the link is down, reconciles on reconnect, presents per-field freshness, and never lets a consumer mistake stale state for live (two-sided safety invariant). It also hosts the beneath-the-agent behavioral-observation vantage. Every other product line rides it; zero-trust, telemetry, and canon are enforced here once and inherited everywhere.

Capabilities

ID	Capability	Role	Tag
CAP-001	Capability Manager (vendor-agnostic facade)	primary	KNOWN
CAP-002	Prompt + capability caching	primary	KNOWN
CAP-003	Custody and data coherence	primary	KNOWN
CAP-004	Evident-staleness / Freshness Quotient	primary	KNOWN
CAP-005	Forwarding and QoS bus semantics	primary	KNOWN
CAP-007	C-1..C-11 custody envelope	primary	KNOWN
CAP-008	Multi-model routing policy	primary	KNOWN
CAP-009	Air-gap / appliance form factor	primary	KNOWN
CAP-019	Alistair Prime cognitive runtime	primary	EMERGENT
CAP-006	Protocol-aware bus adapters	shared	KNOWN
CAP-048	Behavior-based cyber-agent tuning	shared	EMERGENT
CAP-022	HGC3AE2 governance doctrine	shared	KNOWN
CAP-051	Field-level privilege	shared	EMERGENT

PL-2 — Governed Agent Spine

Substrate: requires substrate (rides PL-1)

Maps to: SJ-06 / NOMOS

Wave: W0 Foundation

Technical approach

The governance backbone. Adopts the OAgents conformance levels, the oagent-core single policy decision point (PDP), and tiered agent contracts, then adds the cyber overlay: two-key gates, autonomy-class matrix (L0-L4), interruption contract, and runtime (not point-in-time) conformance enforcement. Closes the documented 'agentic fitness gap' of static governance frameworks.

Capabilities

ID	Capability	Role	Tag
CAP-010	Two-key governance gate	primary	EMERGENT
CAP-011	Interruption contract	primary	EMERGENT
CAP-012	Autonomy-class matrix (L0-L4)	primary	EMERGENT

ID	Capability	Role	Tag
CAP-016	Surface taxonomy (a/f/c)	primary	EMERGENT
CAP-022	HGC3AE2 doctrine	primary	KNOWN
CAP-025	Owned scheduling	primary	EMERGENT
CAP-030	Conformance levels	primary	EMERGENT
CAP-031	Control registry (6-cat)	primary	EMERGENT
CAP-034	6-layer + policy PDP	primary	EMERGENT
CAP-035	Agent contract (6-section tiered)	primary	EMERGENT
CAP-047	Tiered canon	primary	EMERGENT
CAP-032	AICP (shared)	shared	EMERGENT
CAP-036	Cross-model QA (shared)	shared	EMERGENT
CAP-044	Attestation (shared)	shared	KNOWN

PL-3 — Agentic ICAM / Identity

Substrate: requires substrate (rides PL-1)

Maps to: SJ-E-08 / SYMBOLON

Wave: W0 Foundation

Technical approach

Zero-trust agent identity. Productizes the AICP identity-card protocol (per-agent, phase-gated, least-privilege, offline-verifiable, card-multiplexed across domains), the function-scoped credential mesh, and the JIT + observed + field-level privilege controls. Conforms to NIST/FIDO/W3C-DID/A2A rather than competing with them.

Capabilities

ID	Capability	Role	Tag
CAP-014	Function-scoped credential mesh	primary	EMERGENT
CAP-028	Persona charters + per-persona keys	primary	EMERGENT
CAP-032	Agent Identity Card Protocol (AICP)	primary	EMERGENT
CAP-049	Dynamic JIT privilege grant/revoke	primary	EMERGENT
CAP-051	Field-level privilege	primary	EMERGENT
CAP-034	Policy PDP (shared)	shared	EMERGENT
CAP-050	Observed-privilege dashboard (shared)	shared	EMERGENT

PL-4 — Agent Memory & Knowledge Fabric

Substrate: requires substrate (rides PL-1)

Maps to: SJ-E-10 / THESAURUS

Wave: W2 Integrate

Technical approach

Memory-as-a-service. Adopts the Nous three-tier hybrid memory (Director / worker-shared / worker-private; PostgreSQL + per-worker SQLite + embedded vectors) plus git-backed long-term sync. Crowded market (Mem0/Zep/Letta) - the wedge is substrate integration and zero-trust, not memory as a standalone bet.

Capabilities

ID	Capability	Role	Tag
CAP-033	Three-tier hybrid vector memory	primary	EMERGENT
CAP-042	MCP-first PKM / memory-as-a-service	primary	EMERGENT
CAP-046	Git-backed long-term memory sync	primary	EMERGENT

PL-5 — Attestation, Telemetry & Provenance Ledger

Substrate: requires substrate (rides PL-1) **Maps to:** SJ-E-02 / ACTA **Wave:** W0 Foundation

Technical approach

Every action emits a durable, externally-stored event the actor cannot rewrite. Adopts in-toto / SLSA / Sigstore and extends build-grade attestation to agent actions; provenance is produced inside the trust boundary via the substrate vantage. Includes the immutable-artifact + Zenodo DOI publishing pipeline.

Capabilities

ID	Capability	Role	Tag
CAP-013	op-attest telemetry	primary	EMERGENT
CAP-021	Zenodo DOI publishing pipeline	primary	EMERGENT
CAP-029	Additive-doctrine immutable artifacts	primary	EMERGENT
CAP-041	Zenodo MCP publisher	primary	EMERGENT
CAP-044	Attestation (in-toto/SLSA/SCITT/RATS)	primary	KNOWN
CAP-004	Freshness Quotient (shared)	shared	KNOWN

PL-6 — Delivery & Systems Engineering

Substrate: requires substrate (rides PL-1) **Maps to:** SJ-04 / FABRICA **Wave:** W2 Integrate

Technical approach

The governed delivery engine. SAFe/Scrum decomposition and triage pipelines, the MBSE PLM phase state machine (eidolon), .ologic visual modeling (Ordinal), and the editorial 8-stage state machine for long-form authored work.

Capabilities

ID	Capability	Role	Tag
CAP-017	Decomposition pipeline	primary	EMERGENT
CAP-018	Pill triage workflows	primary	EMERGENT
CAP-020	Editorial 8-stage state machine	primary	EMERGENT
CAP-039	.ologic visual modeling	primary	EMERGENT
CAP-040	PLM phase state machine	primary	EMERGENT

PL-7 — Ops Platform & Console

Substrate: requires substrate (rides PL-1) **Maps to:** SJ-02 / ARX (+ OCULUS dashboard) **Wave:** W0/W2

Technical approach

The operating plane. Platform-OS deploy via OIDC->SSM zero-static-credential federation, K3s and systemd-Docker on Debian, ops-data centralization, the agent-state console, and - shipped in W0 - the OCULUS observed-privilege real-time dashboard (the substrate never ships without it).

Capabilities

ID	Capability	Role	Tag
CAP-015	OIDC->SSM zero-static-cred deploy	primary	EMERGENT
CAP-023	Ops-data centralization	primary	EMERGENT
CAP-024	K3s deploy	primary	EMERGENT
CAP-027	Ops listener bot	primary	EMERGENT
CAP-043	systemd-Docker on Debian	primary	EMERGENT
CAP-045	Agent-state console + backlog viewer	primary	EMERGENT
CAP-050	Observed-privilege dashboard	primary	EMERGENT
CAP-034	Policy PDP (shared)	shared	EMERGENT
CAP-051	Field-level privilege (shared)	shared	EMERGENT

PL-8 — Agent Behavioral Observability & Tuning

Substrate: requires substrate (rides PL-1) **Maps to:** SJ-E-09 / NEMESIS **Wave:** W1 Lead

Technical approach

Agent-EDR. Observes each agent from beneath at the substrate (ground truth, not self-report), detects drift and injection-driven misbehavior, and tunes or constrains the agent in-flight. The structural antidote to self-attestation; the runtime detection-and-correction layer the cascade literature shows is missing.

Capabilities

ID	Capability	Role	Tag
CAP-048	Behavior-based cyber-agent tuning	primary	EMERGENT
CAP-004	Freshness Quotient (shared)	shared	KNOWN
CAP-013	op-attest telemetry (shared)	shared	EMERGENT
CAP-034	Policy PDP (shared)	shared	EMERGENT
CAP-032	AICP (shared)	shared	EMERGENT
CAP-050	Observed-privilege (shared)	shared	EMERGENT

PL-9 — Agent Assurance / QA & Eval

Substrate: requires substrate (rides PL-1) **Maps to:** SJ-E-11 / EXAMEN **Wave:** W2 Integrate

Technical approach

Independent verification. Adopts the o-qa 8-dimension cross-model QA engine and the evaluation-harness-as-conformance-engine; verifier != author, on separate instances. Directly enforces the no-self-attestation rule (Zero-Trust Law 1).

Capabilities

ID	Capability	Role	Tag
CAP-026	Klaus integrity / QA sweeps	primary	EMERGENT
CAP-036	8-dimension cross-model QA	primary	EMERGENT
CAP-037	Eval harness as conformance engine	primary	EMERGENT
CAP-038	Zero-dependency portable skills	primary	EMERGENT
CAP-048	Behavioral tuning (shared)	shared	EMERGENT

PL-10 — OT/Edge Advisory Sidecar

Substrate: requires substrate (rides PL-1) **Maps to:** SJ-E-03 / MENTOR **Wave:** W3 Horizon

Technical approach

Industrial advisory. Speaks the OT/integration buses natively (OPC-UA, MQTT/Sparkplug, Modbus) as a sidecar at the advisory/analytics tier - beside the control loop, never inside it, with no real-time control writes by default. Per-tag freshness so a cached sensor value is never mistaken for live.

Capabilities

ID	Capability	Role	Tag
CAP-006	Protocol-aware bus adapters	primary	KNOWN
CAP-003	Custody and data coherence (shared)	shared	KNOWN
CAP-004	Freshness Quotient (shared)	shared	KNOWN
CAP-005	QoS bus semantics (shared)	shared	KNOWN